THE UNIVERSITY of EDINBURGH

USHER MASTERS RESEARCH ETHICS GROUP
email: UMREG@ed.ac.uk

10 December 2025

Certification of Institutional Favourable Ethical Opinion

The Usher Masters Research Ethics Group (UMREG), Usher Institute, University of Edinburgh has reviewed and given a favourable assessment to the following Masters level student research project:

Ethics ID: UM251145

Principal Investigator: Benjamin Gross

Supervisor(s): Dr. Syed Ahmar Shah, John Wilson

Collaborators(s):

Study Title: Detecting Movement Imbalances using a custom TensorFlow Lite Model utilizing Google MediaPipe framework.

Sponsor (if applicable):

Sponsor Number (if applicable):

The UMREG operates in accordance with the guidelines set out by the Edinburgh Medical Research Ethics Committee (EMREC).

Restrictions:

This certification is subject to the following conditions:

- Favourable ethical opinion is granted only for the research and purposes described in the application.
- Any modification to the approved research must be submitted to UMREG for approval

We wish you the best of luck with your research.

Approved by: **Date:** 10 December 2025

Su Goopy, Lead, UMREG